

Modelling and Control of DC Motor

1. Mathematical Model of DC Motor

The goal in the development of the mathematical model is to relate the voltage applied to the armature to the velocity of the motor. Two balance equations can be developed by considering the electrical and mechanical characteristics of the system.

1.1 Electrical Characteristics

The equivalent electrical circuit of a dc motor is illustrated in figure1. It can be represented by a voltage source (V_a) across the coil of the armature. The electrical equivalent of the armature coil can be described by an inductance (L_a) in series with a resistance (R_a) in series with an induced voltage (V_c) which opposes the voltage source. The induced voltage is generated by the rotation of the electrical coil through the fixed flux lines of the permanent magnets. This voltage is often referred to as the back *emf* (electromotive force).

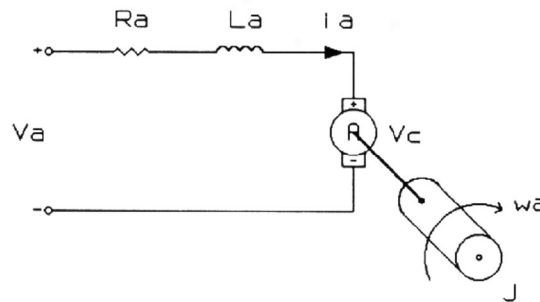


Figure 1: Representation of DC Motor

A differential equation for the equivalent circuit can be derived by using Kirchhoff's voltage law around the electrical loop. Kirchhoff's voltage law states that the sum of all voltages around a loop must equal zero, or

$$V_a - V_{Ra} - V_{La} - V_c = 0 \quad (1)$$

According to Ohm's law, the voltage across the resistor can be represented as

$$V_{Ra} = i_a R_a \quad (2)$$

where i_a is the armature current. The voltage across the inductor is proportional to the change of current through the coil with respect to time and can be written as

$$V_{La} = L_a \frac{di_a}{dt} \quad (3)$$

where L_a is the inductance of the armature coil. Finally, the back *emf* can be written as

$$V_c = K_v \omega_a \quad (4)$$

where K_v is the velocity constant determined by the flux density of the permanent magnets, the reluctance of the iron core of the armature, and the number of turns of the armature winding. ω_a is the rotational velocity of the armature.

Substituting all equations (1-4) can be get the following differential equation

$$V_a - i_a R_a - L_a \frac{di_a}{dt} - K_v \omega_a = 0 \quad (5)$$

1.2 Mechanical Characteristics

Performing an energy balance on the system, the sum of the torques of the motor must equal zero. Therefore,

$$T_e - T_{\omega 1} - T_{\omega} - T_L = 0 \quad (6)$$

where T_e is the electromagnetic torque, $T_{\omega 1}$ is the torque due to rotational acceleration of the rotor, T_{ω} is the torque produced from the velocity of the rotor, and T_L is the torque of the mechanical load. The electromagnetic torque is proportional to the current through the armature winding and can be written as

$$T_e = K_t i_a \quad (7)$$

where K_t is the torque constant and like the velocity constant is dependent on the flux density of the fixed magnets, the reluctance of the iron core, and the number of turns in the armature winding. $T_{\omega 1}$ can be written as

$$T_{\omega 1} = J \frac{d\omega_a}{dt} \quad (8)$$

where J is the inertia of the rotor and the equivalent mechanical load. The torque associated with the velocity is written as

$$T_{\omega} = B \omega_a \quad (9)$$

where B is the damping coefficient associated with the mechanical rotational system of the machine.

Substituting all equations (6-9) can be get the following differential equation

$$K_t i_a - J \frac{d\omega_a}{dt} - B\omega_a - T_L = 0 \quad (10)$$

1.3 Transfer Function Block Diagram

A block diagram for the system can be developed from the differential equations given in eqns. (5) and (10). Taking the Laplace transform of each equation gives

$$sI_a(s) - i_a(0) = -\frac{R_a}{L_a}I_a(s) - \frac{K_v}{L_a}\Omega_a(s) + \frac{1}{L_a}V_a(s)$$

$$s\Omega_a(s) - \omega_a(0) = \frac{K_t}{J}I_a(s) - \frac{B}{J}\Omega_a(s) - \frac{1}{J}T_L(s)$$

The above equations can then easily be put into block diagram form. The block diagram obtained from these equations for a permanent magnet dc motor is shown in figure 2.

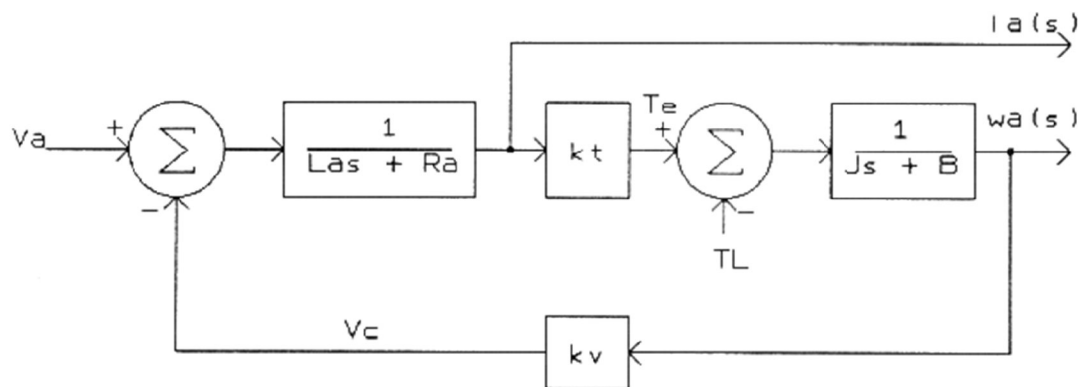


Figure 2: Block Diagram of DC Motor

2. Modeling of DC Motor Using Simulink

To simulate the DC Motor using Simulink can be use the transfer function block (**Transfer Fcn**), and build the same block diagram in figure 2 (**Open Loop Model**)

- 1) Go to **Simulink** > **Continuous**; Insert the (**Transfer Fcn**).
- 2) Go to **Simulink** > **Commonly Used Blocks**; Insert the (**Sum**). And (**Gain**).

Assuming the DC Motor has the following parameter

Armature Resistance (R_a)	$R_a = 0.6324\Omega$
Armature Inductance (L_a)	$L_a = 0.9134 H$
Back emf Constant (K_v)	$K_v = 0.1270 \frac{V}{rad/sec}$
Torque Constant (K_t)	$K_t = 0.1270 \frac{N}{A}$
Mechanical Inertia (J)	$J = 0.9058 kg.m^2$
Friction Coefficient (B)	$B = 0.8147 \frac{N.m}{rad/sec}$

The Simulink block will appear as figure 3

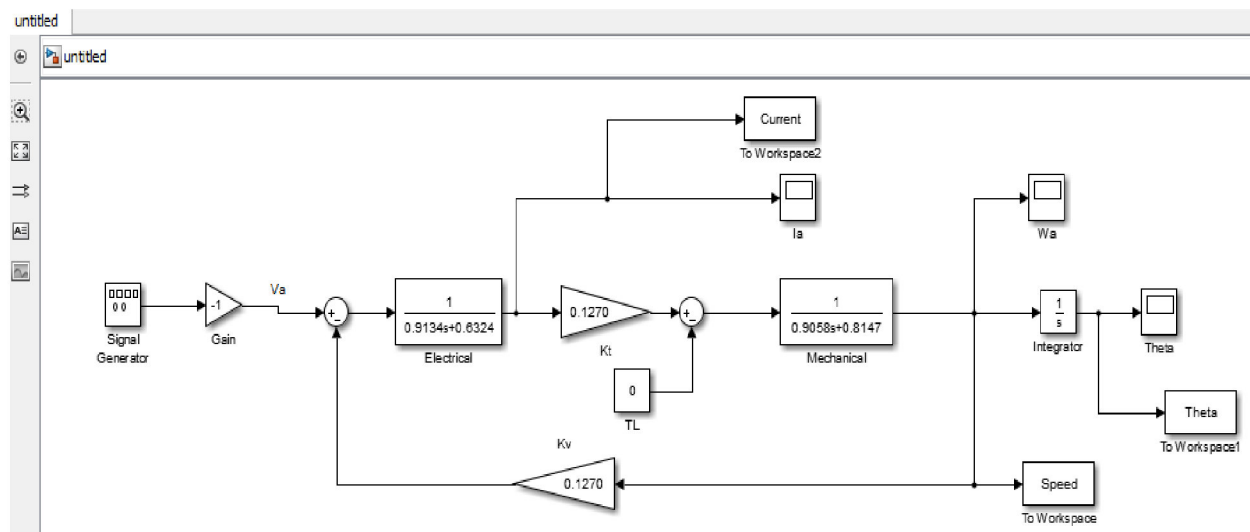


Figure 3: Simulink Model of DC Motor

Assume the **Signal Generator** ; choose the wave form square with **Amplitude 5 and Frequency 1/50 Hz**.

The result of speed , angle and current is shown in the following figures

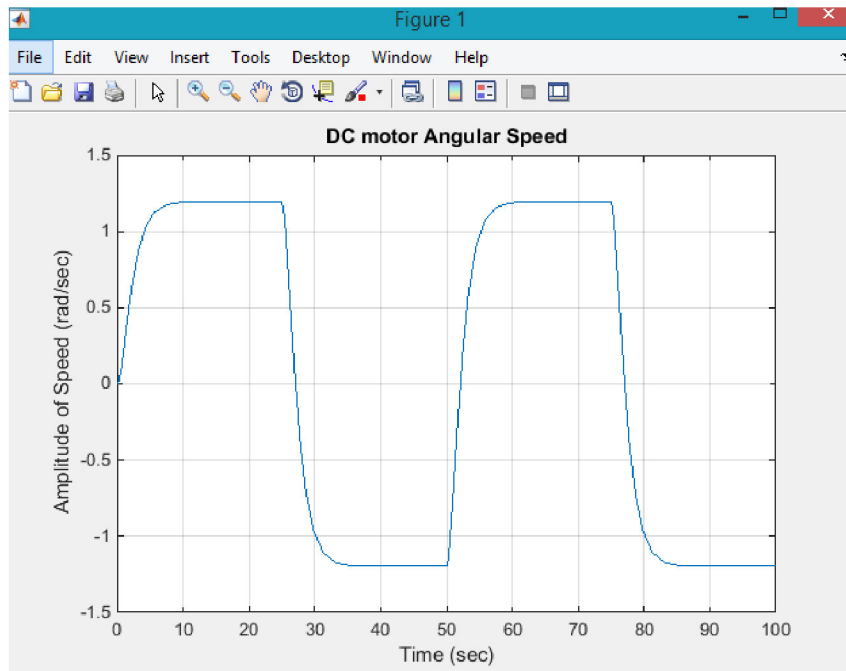


Figure 4: Speed of DC Motor

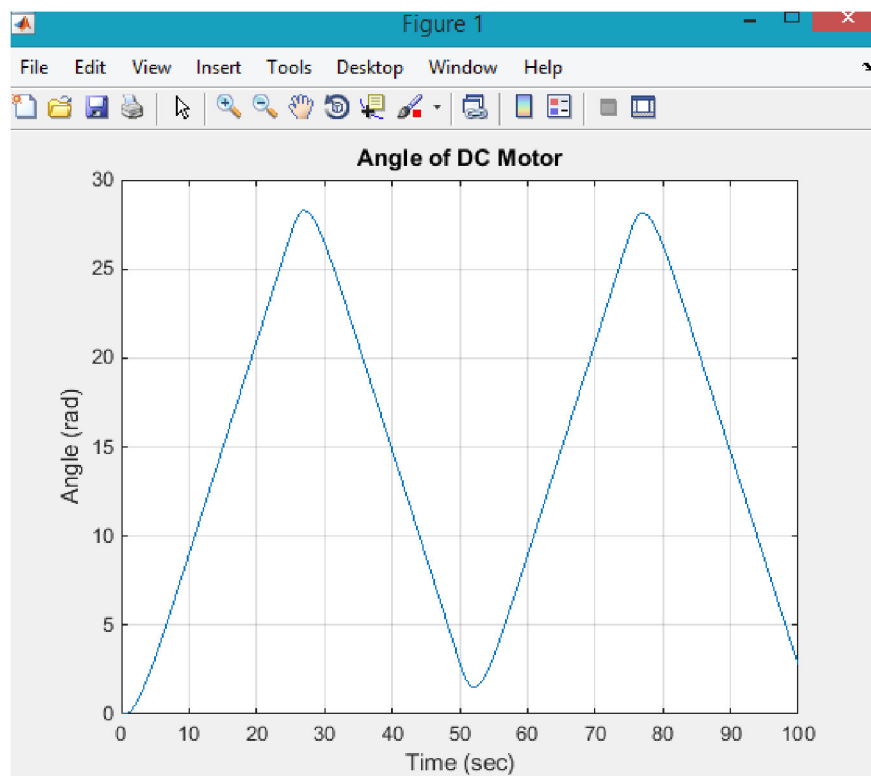


Figure 5: Angle of DC Motor

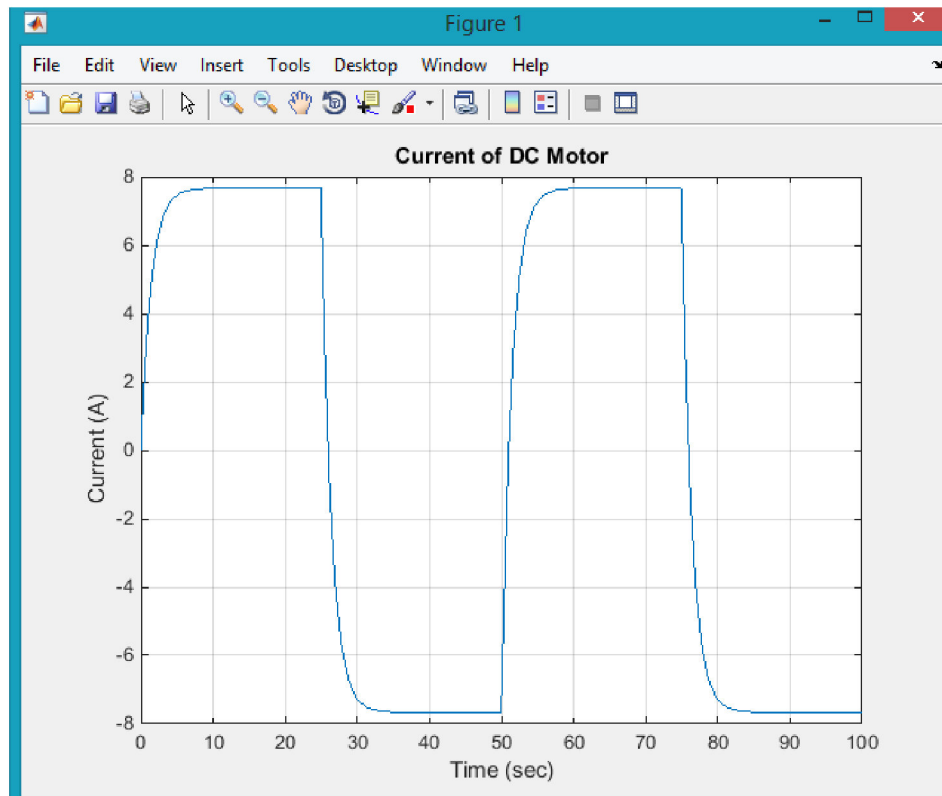


Figure 6: Current of DC Motor

3. Modeling of DC Motor using SimScape (Open Loop System)

The Mathematical model of DC motor will contain two part the electrical circuit and the mechanical circuit. To be able connect the two circuit together should be a converter between mechanical and electrical elements. The model of DC motor can be construct by the following steps:

- 1) Go to *Simscape > Foundation Library > Electrical > Electrical Elements*
Choose the blocks (**Resistor, Inductor and Rotational Electromechanical Converter**).
- 2) Go to *Simscape > Foundation Library > Electrical > Electrical Sensors*
Choose **Current Sensor**.
- 3) Go to *Simscape > Foundation Library > Electrical > Electrical Sources*
Choose **Voltage Controlled Source**
- 4) Go to *Simscape > Foundation Library > Mechanical > Rotational Mechanical Elements*
Choose the following block (**Inertia and Damper**)
- 5) Go to *Simscape > Foundation Library > Mechanical > Mechanical Sensors*
Choose **Ideal Rotational Motion Sensor**.
- 6) Go to *Simscape > Foundation Library > Mechanical > Mechanical Sources*
Choose **Ideal Torque Source**
- 7) Go to *Simscape > Utilities*
Choose the blocks (**PS-Simulink Converter, Simulink-PS Converter and Solver**)
- 8) Connect the all blocks as shown in the figure 7

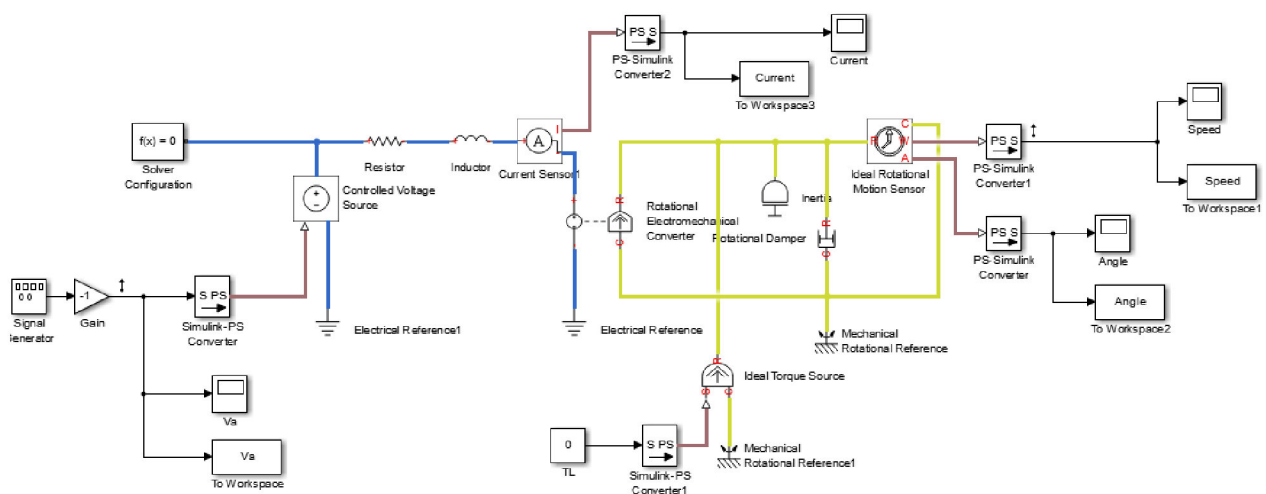
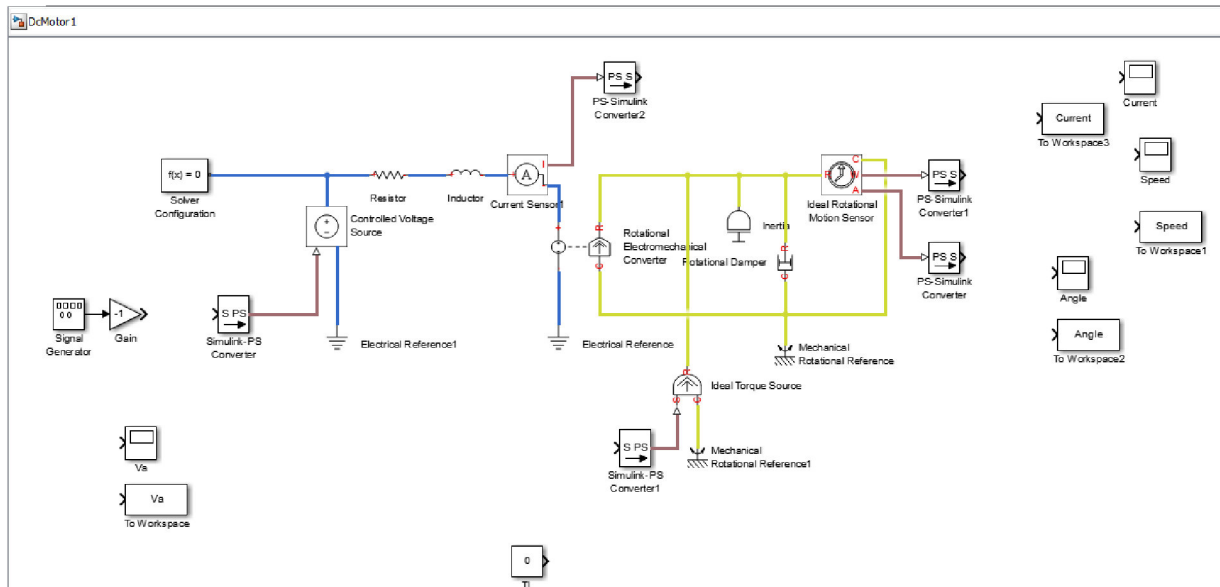


Figure 7: Simscape Model of DC Motor

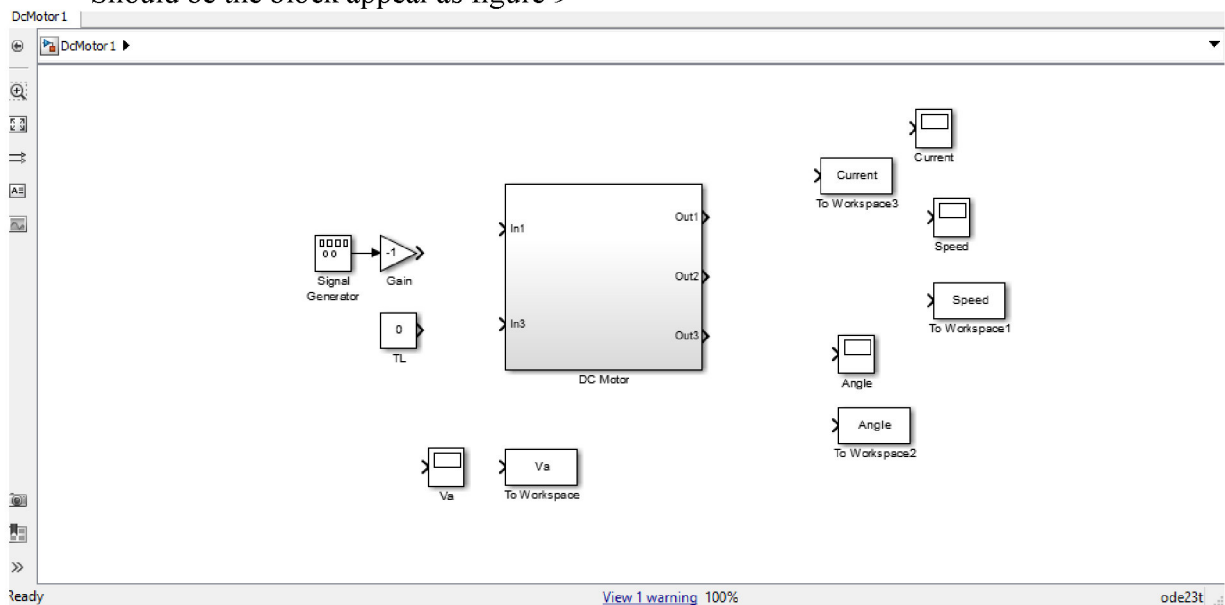
The figure 7 shows the open loop system of DC motor , the simulation result will be the same result that shown in the figures 4, 5 and 6.

To study the closed loop characteristic and response of DC motor , we will make the Simscape model as subsystem by the following steps:

- 1) Remove the connection of scope and input as shown in figure 8



- 2) Select all Simscape block and right click choose **Create subsystem for selection or ctrl+G**
Should be the block appear as figure 9



- 3) Double click on the DC Motor subsystem and change the name of input and output node as figure 10 and figure 11

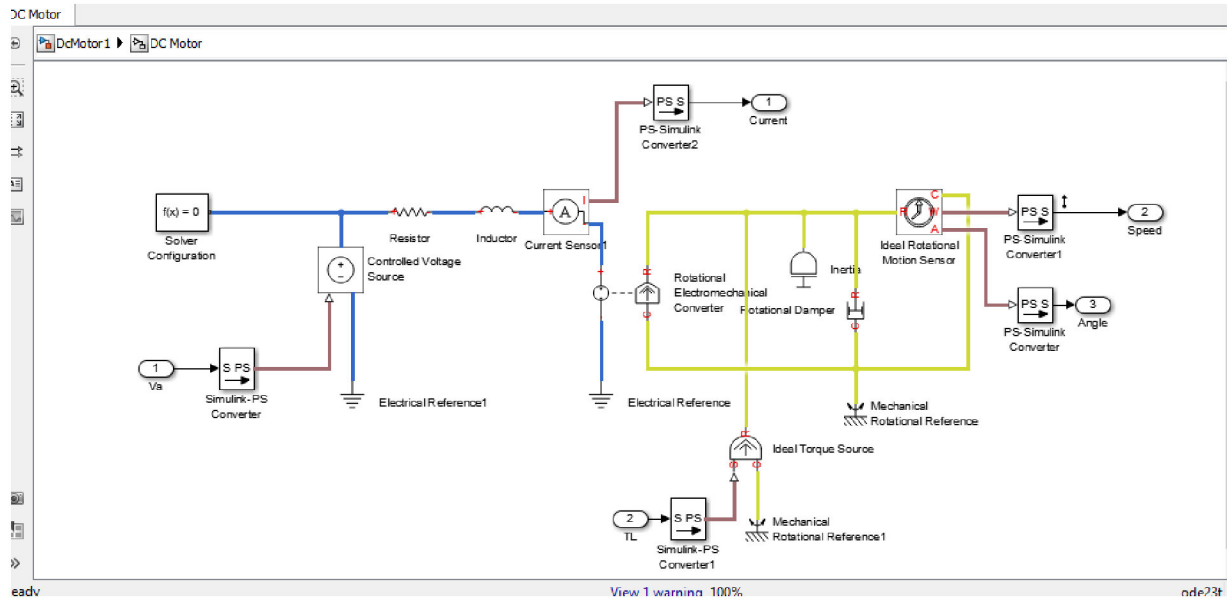


Figure 10

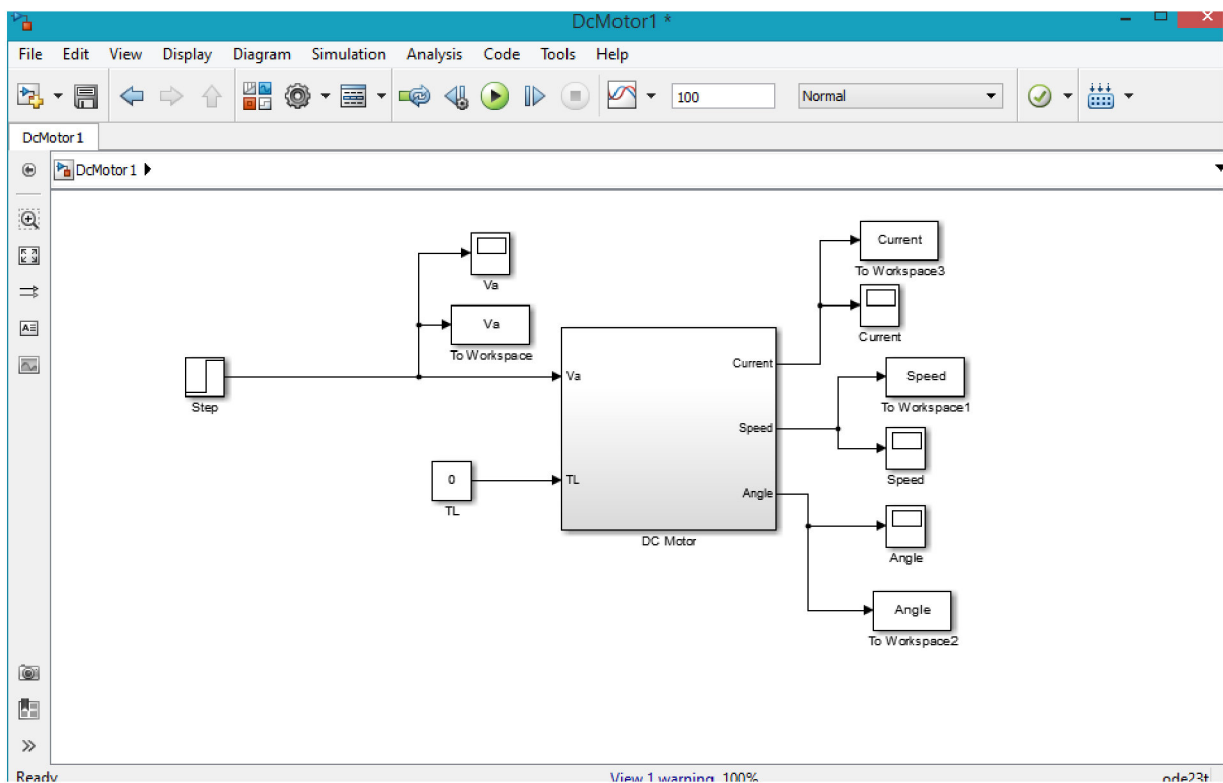
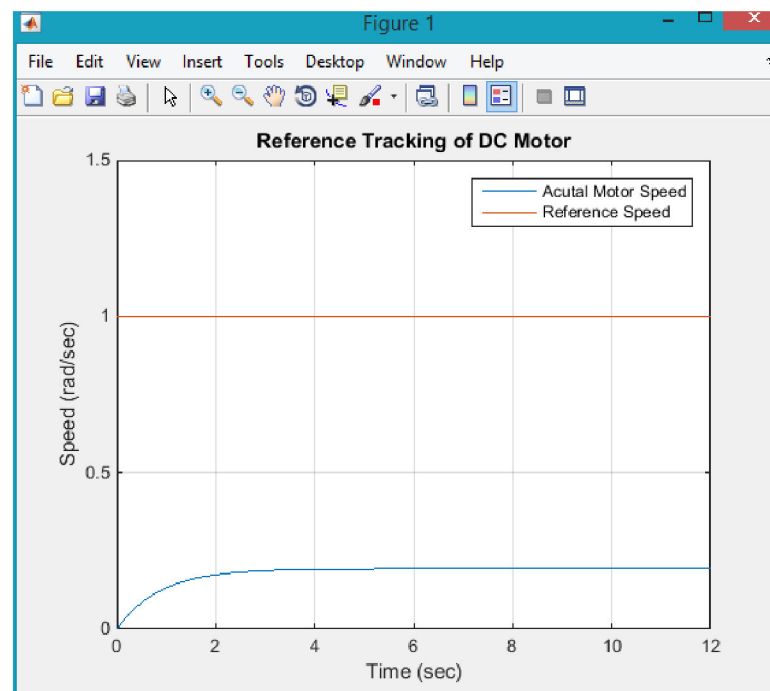
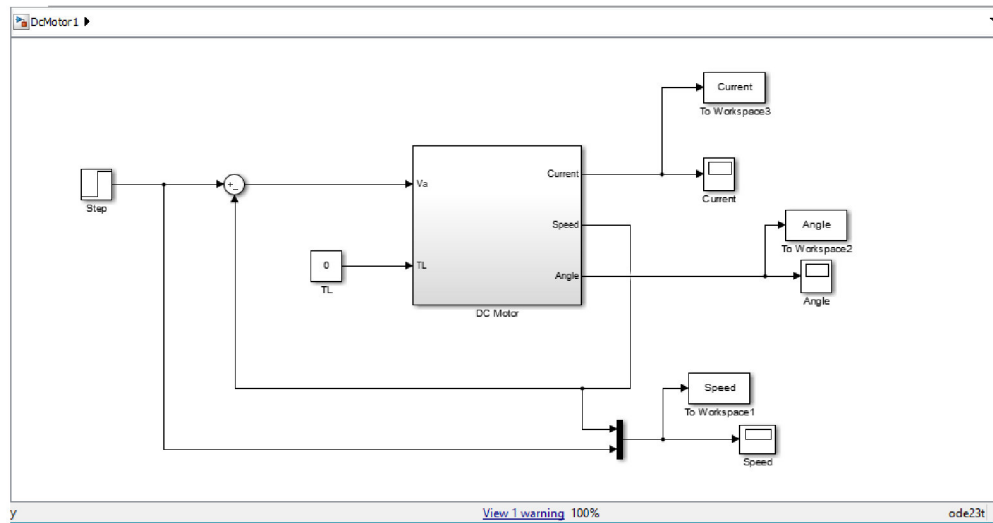


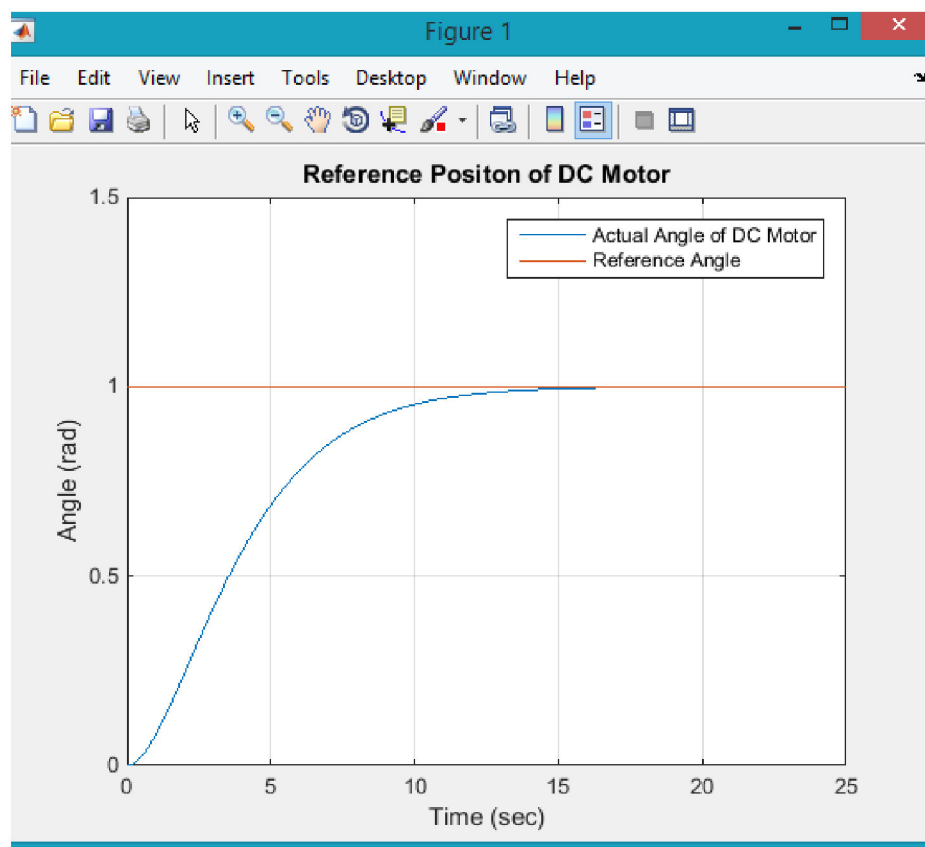
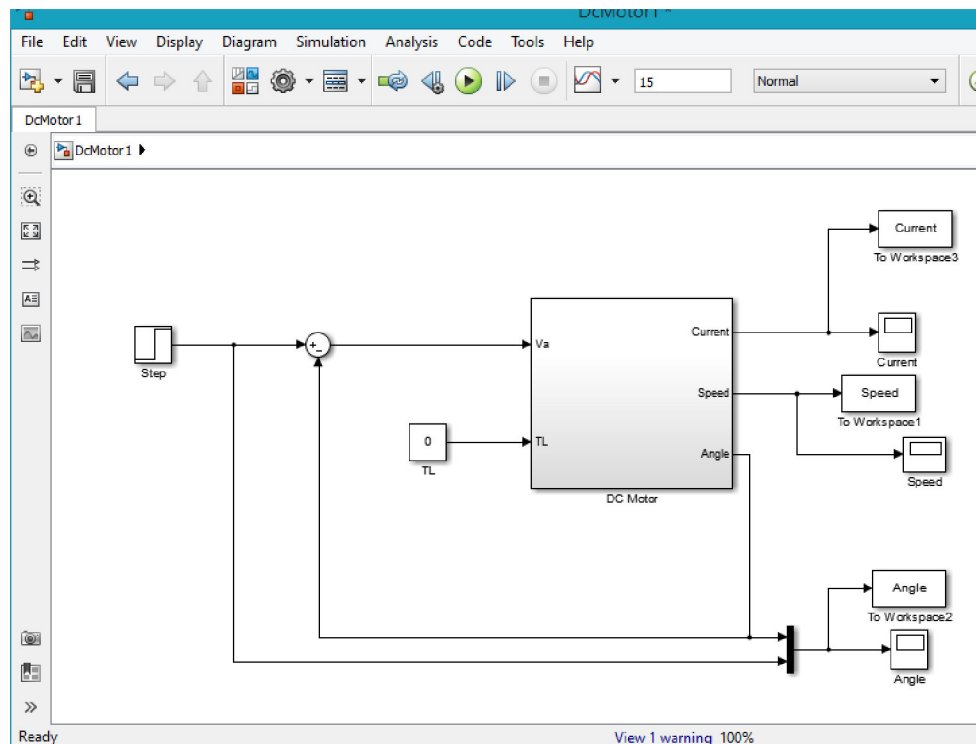
Figure 11

Now we are ready to study the closed loop characteristic of DC Motor for Position and Speed as following

4.1 Closed Loop System (Reference Speed)



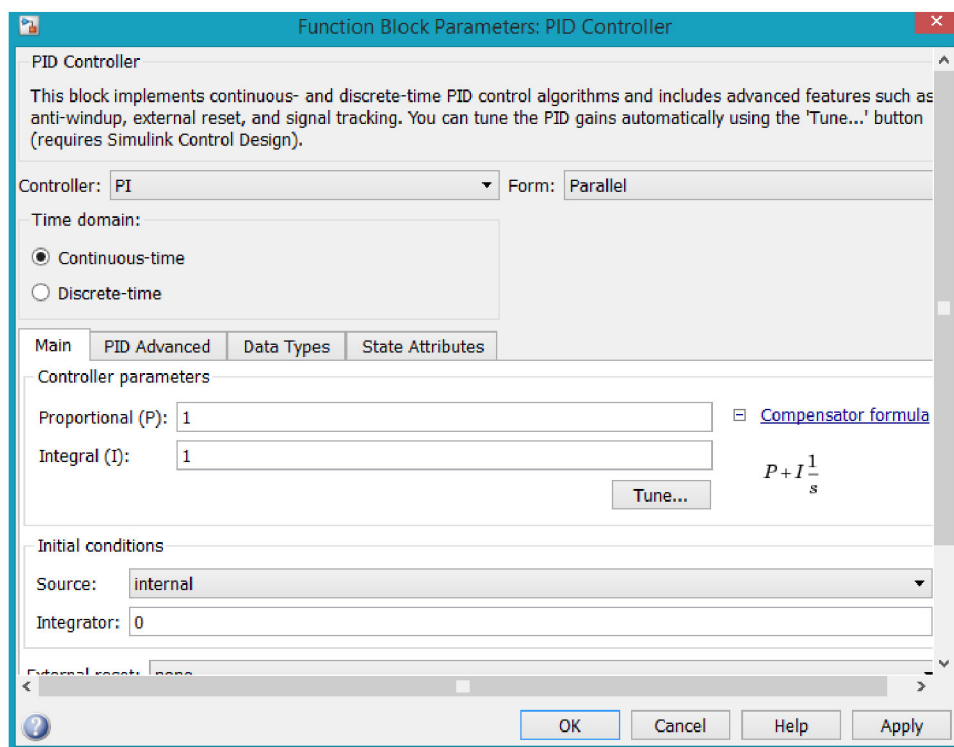
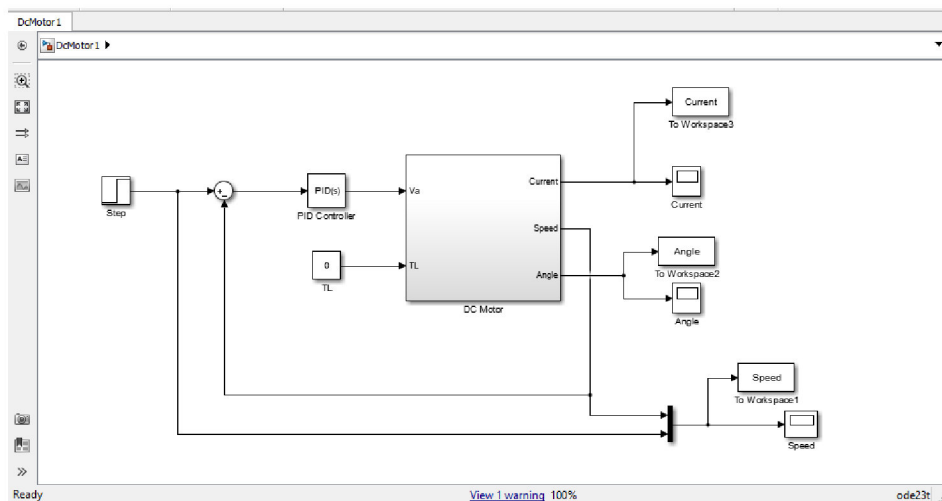
4.2 Closed Loop System (Reference Angle)



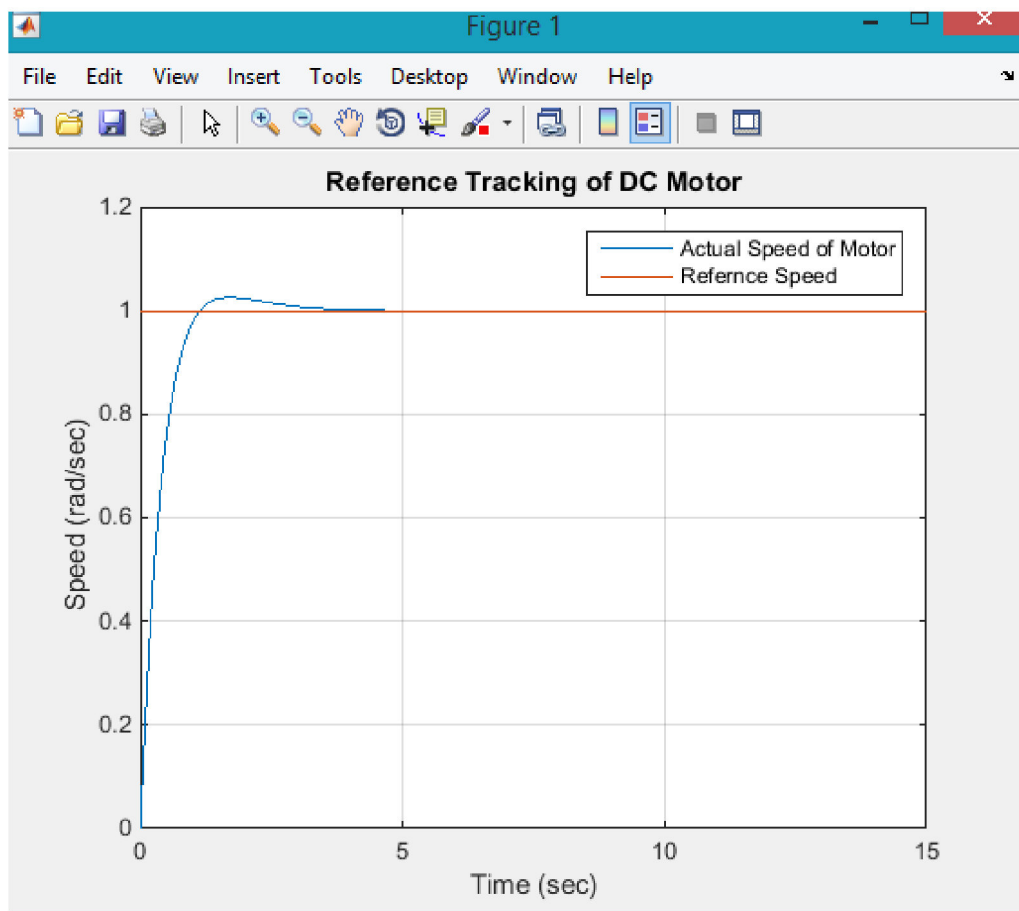
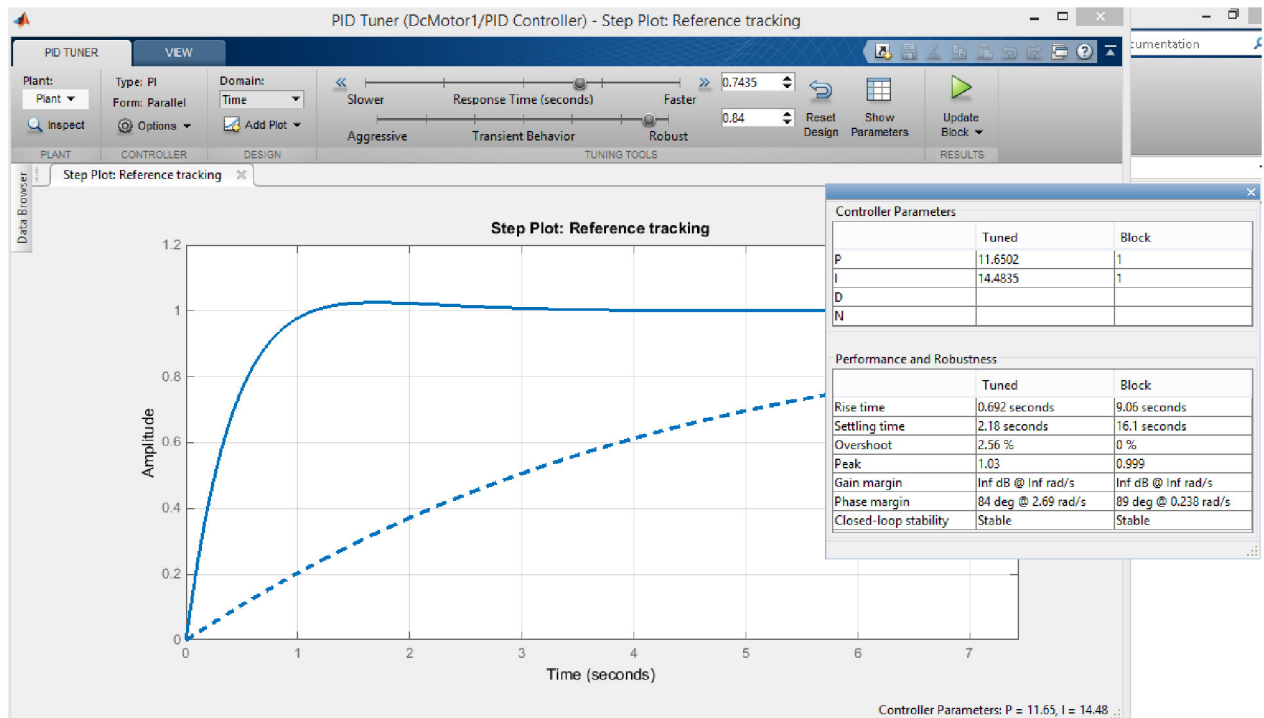
5. Speed and Position Controller for DC Motor

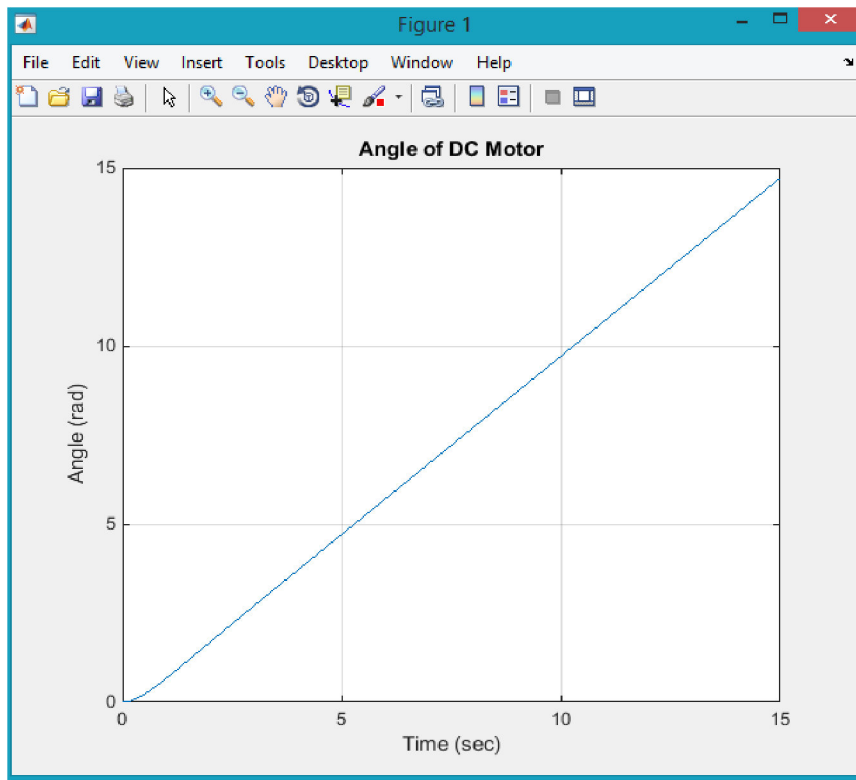
Can be design a PID control using PID tuner toolbox to satisfy the reference tracking of DC motor , in this section we design a PID controller to control the speed of DC motor and PID controller to control the positon of DC motor.

5.1 Speed Control of DC Motor

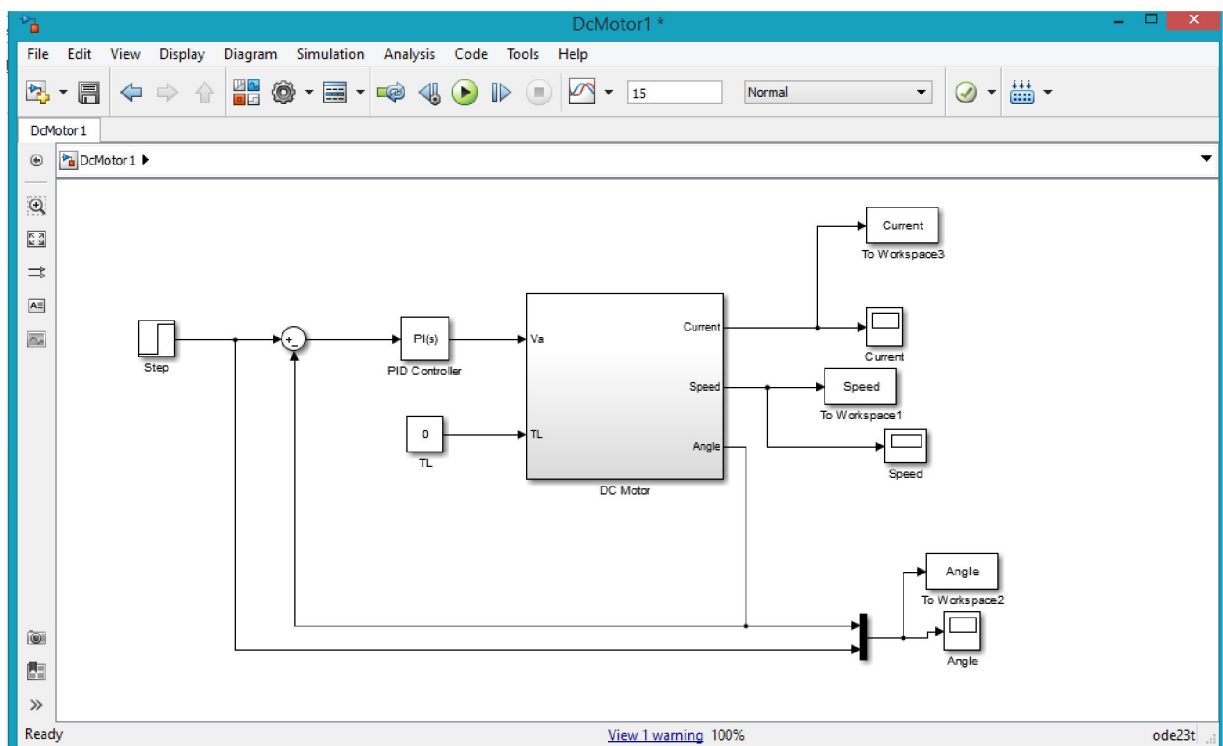


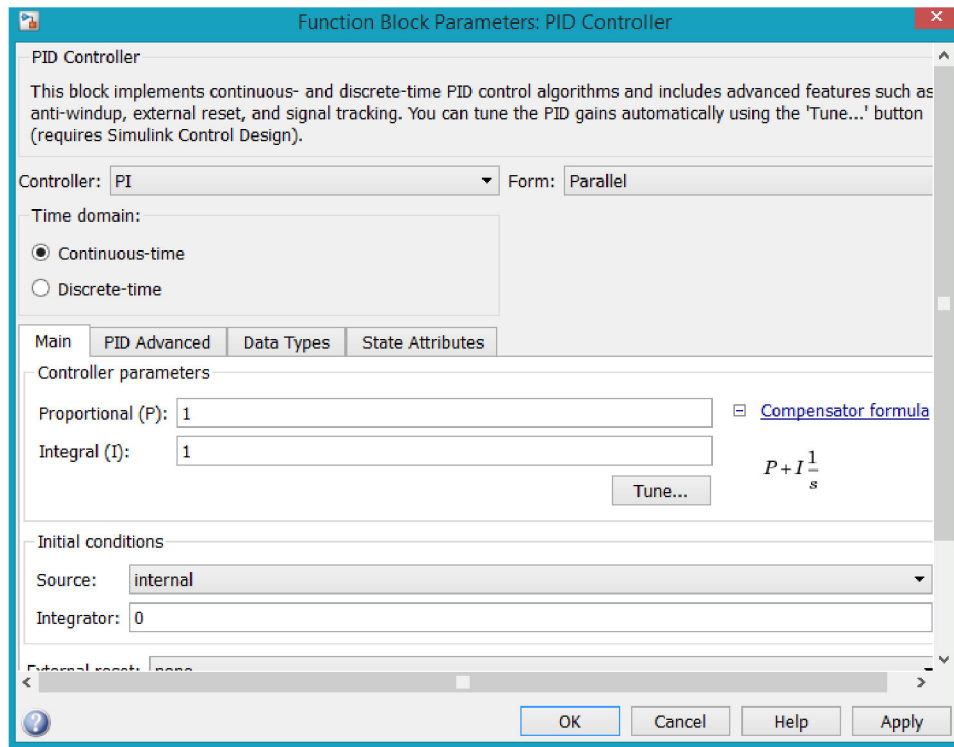
Choose PI controller and press Tune to show the PID tuner toolbox and change the response until satisfy the required specification



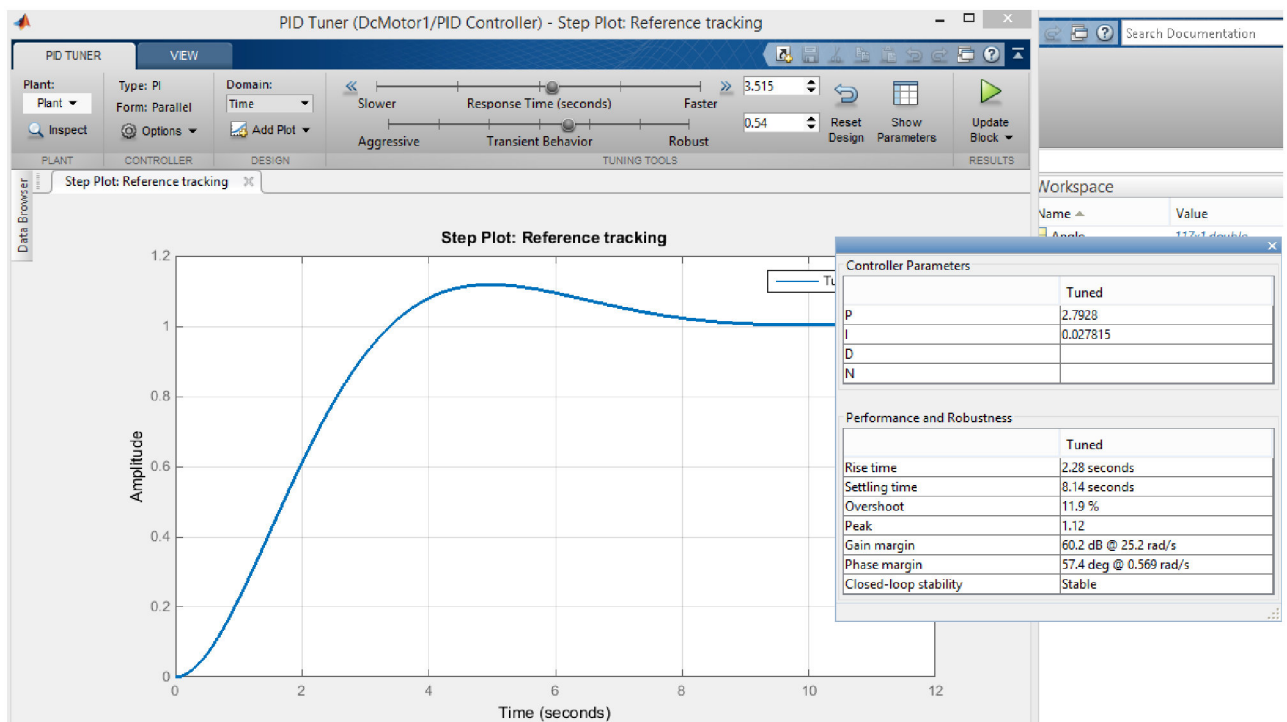


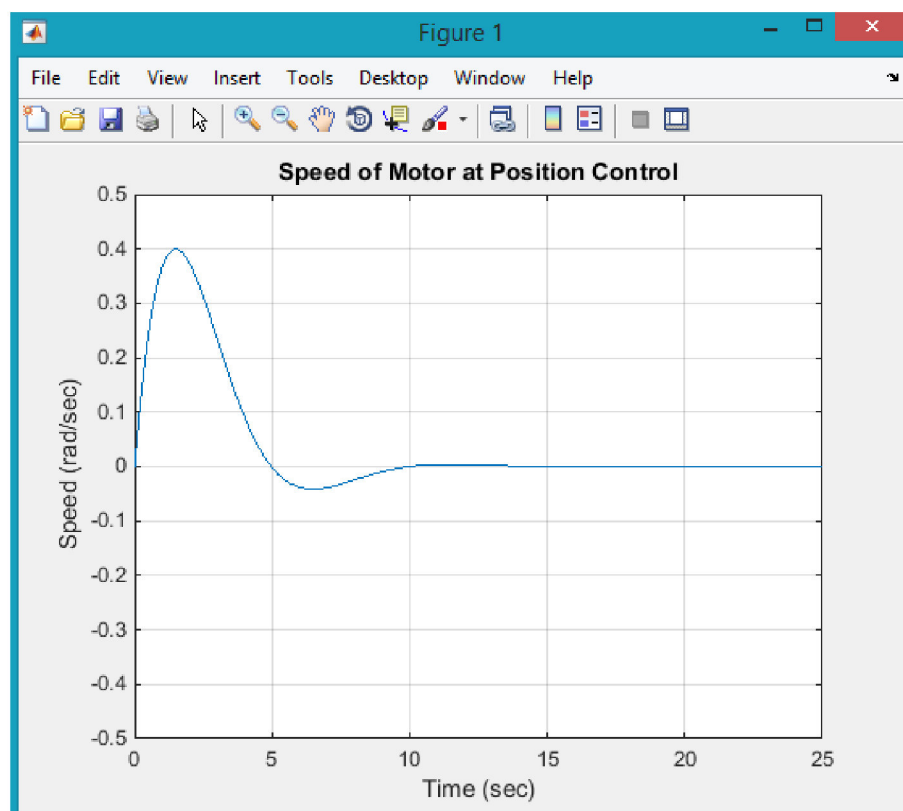
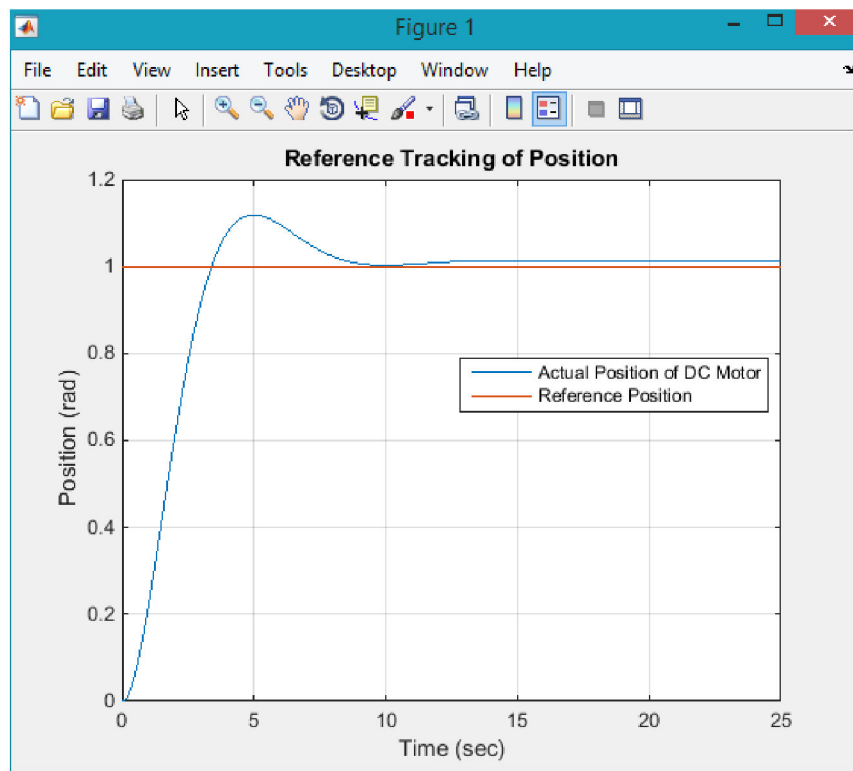
5.2 Position Control of DC Motor





Choose PI controller and press Tune to show the PID tuner toolbox and change the response until satisfy the required specification.





6. System Identification of DC Motor

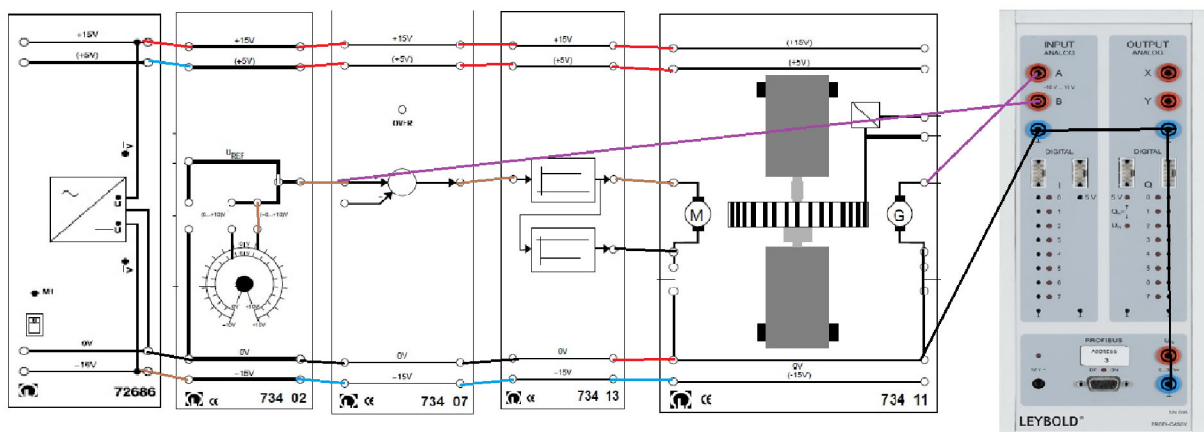
As discussed in the previous sections, we observe that we can build the DC motor using block diagram or using the Simscape model to obtain the simulation result. And the parameter in two cases are known. And the question now is what can be done if we don't have the parameter of the motor??

In this section, we will show how we can get the open loop transfer function of a DC motor using experimental data and the System identification toolbox in MATLAB.

The important part of the identification process is to get the experimental data from the system. In this experiment, the DC motor has two outputs: the speed of the shaft and the position (Angle) of the shaft. We can measure these two values using a tachometer to measure speed and a rotational potentiometer to measure the position.

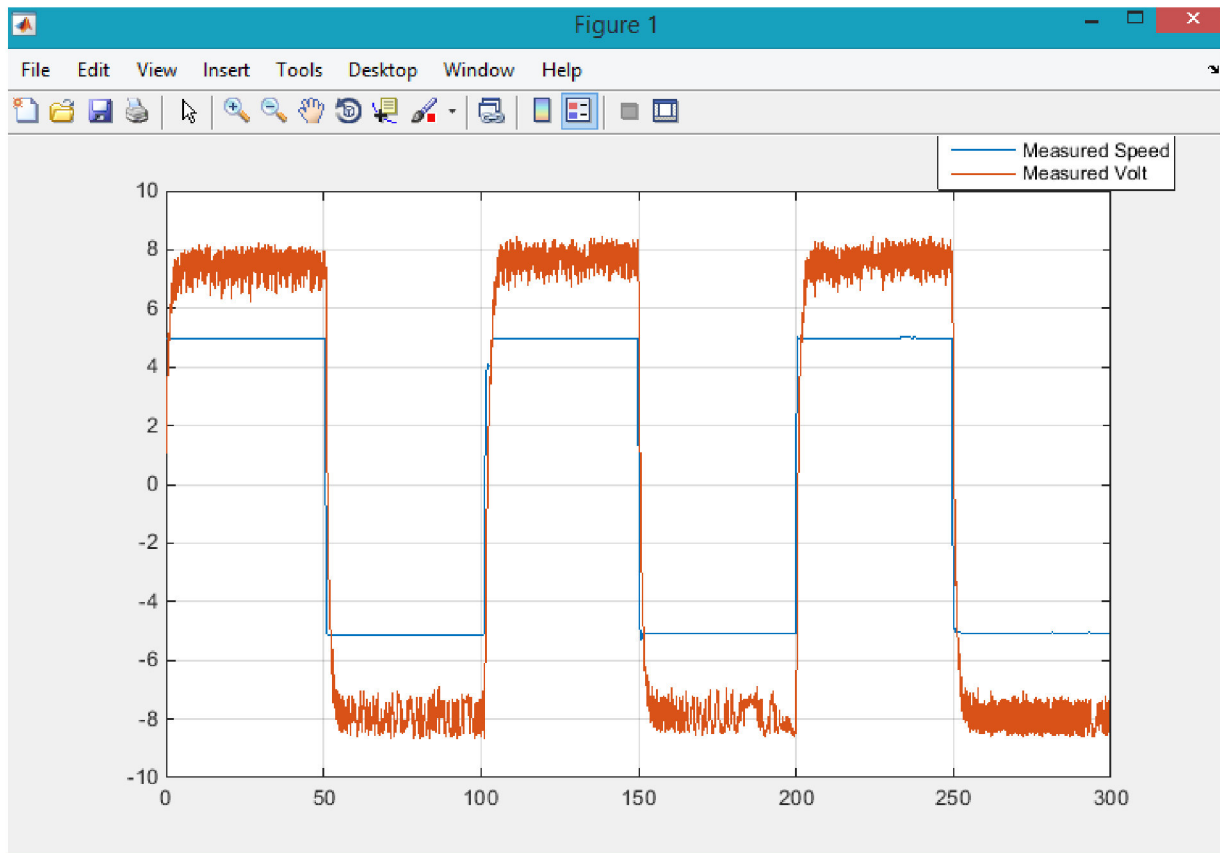
6.1 Measure the Data from DC Motor

To obtain the speed measured data, we will connect the system as shown in the following figure.



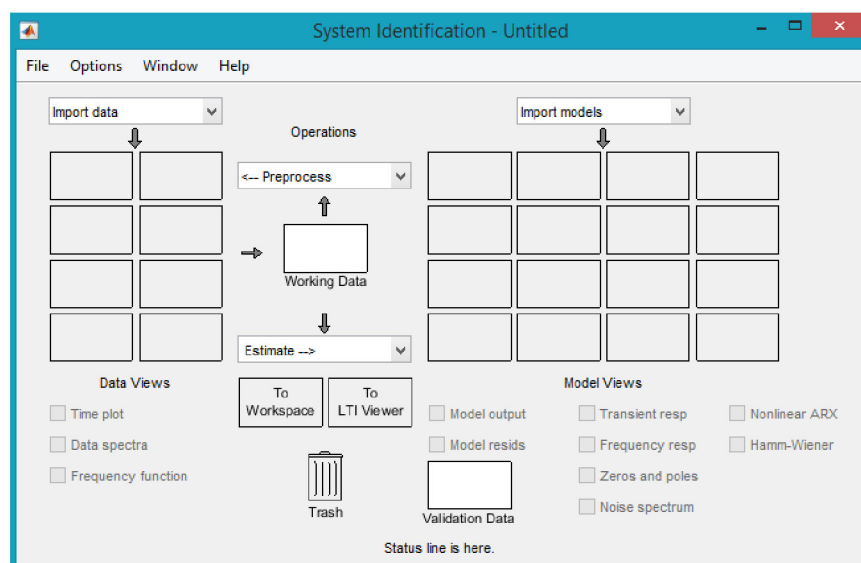
The input of the system will be assumed as a square wave signal with a magnitude of (+5V), and we will use Cassy Lab software to measure the data and transfer it to MATLAB.

The measured speed of the DC motor will be as shown in the following figure, and we will save the Volt, Speed, and time as a **.mat** file.

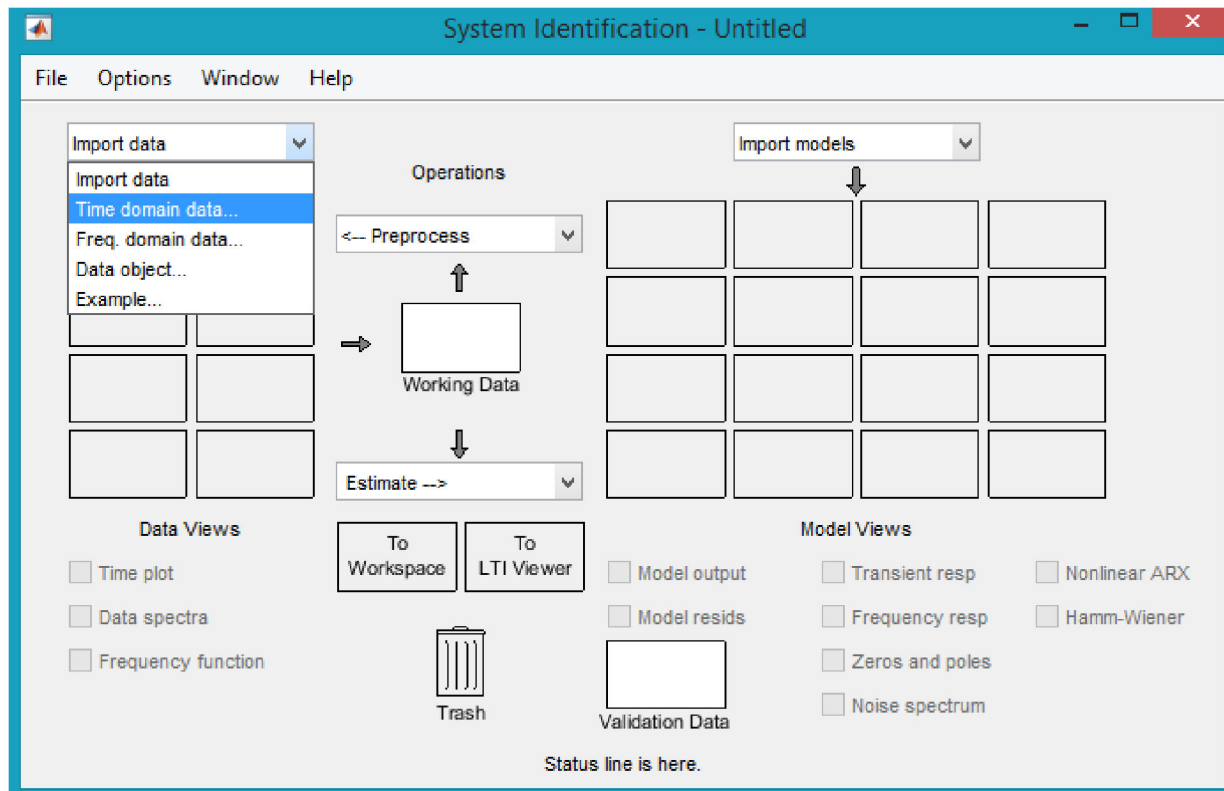


No we take this data and import to system identification toolbox in MATLAB to obtain the transfer function, as the following steps:

- 1) In Command Window of MATLAB write (*ident*); this command to open system identification toolbox.



- 2) The toolbox have the main part (*Import Data, Import Models and Estimate*)
 Import Data : to get the testing and experimental data
 Import Models: show the identification model and its response
 Estimate: to choose the output model form (Transfer Function, State Space, ..etc)
- 3) The first step is import the experimental data form the values stored in the workspace as shown in figure.



There is two types of data: Time Domain Data and Frequency Domain Data, most cases in the system identification the data was used is time domain data.

In this experiment, we have time domain data stored in work space by (Volt, Speed and time).

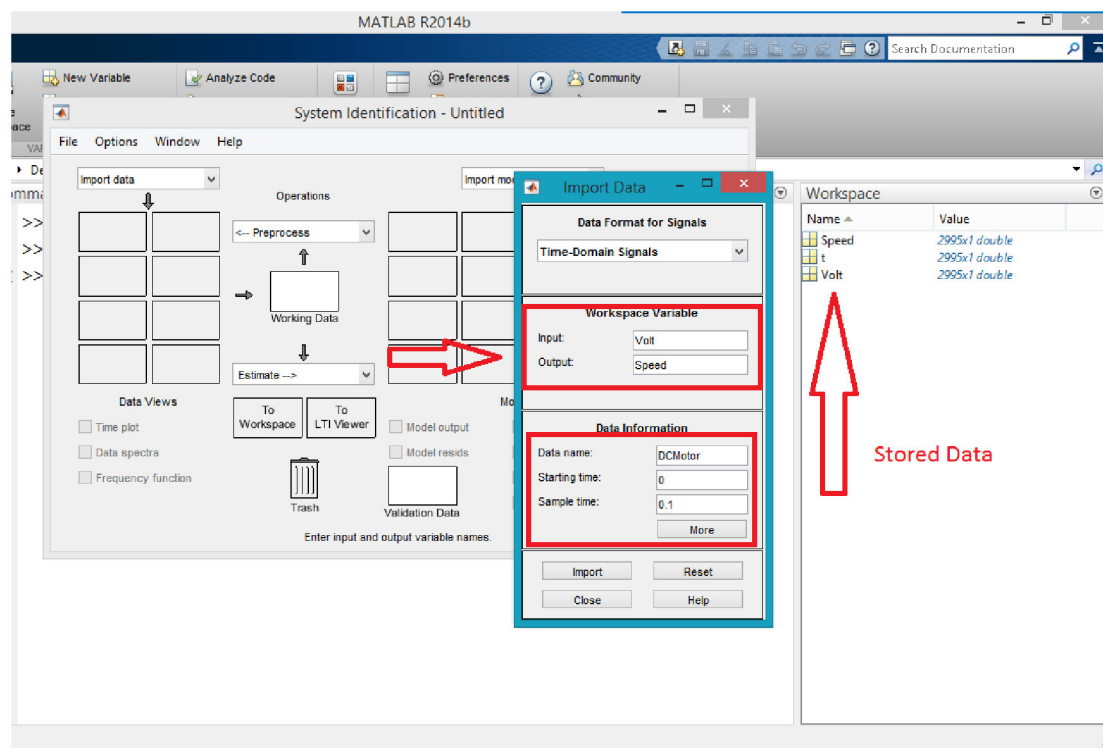
The input data is variable (Volt)

The Output data is variable (Speed)

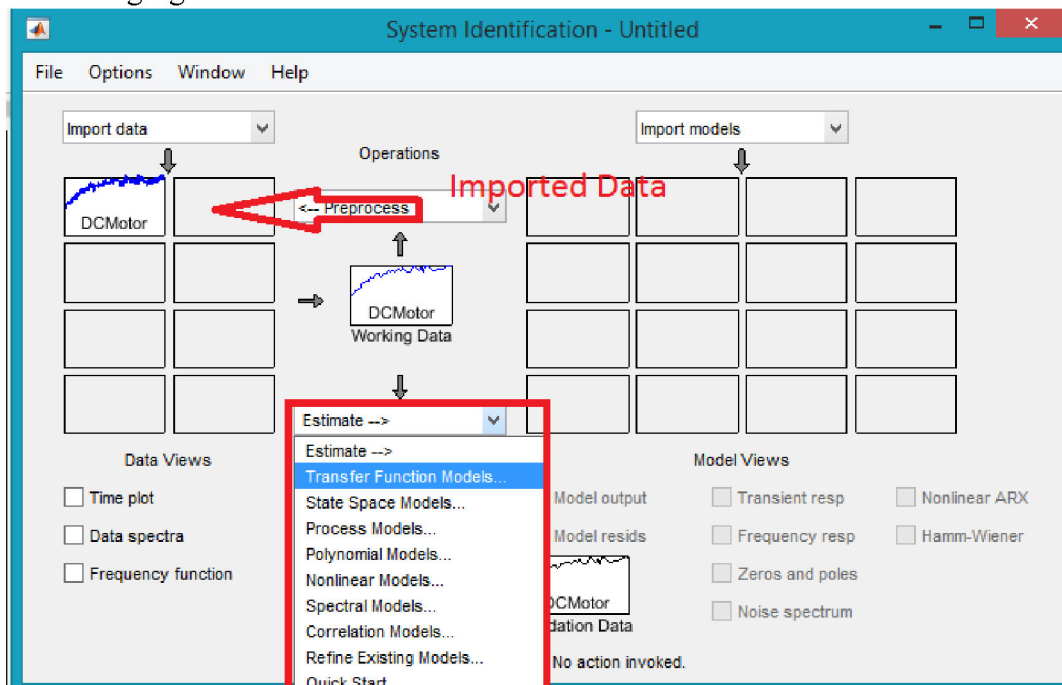
And we set the starting time = 0

And the sample time = 0.1 (the difference between two value in the time data)

Note: the sample time is very important value because can be generate the model with different specification.

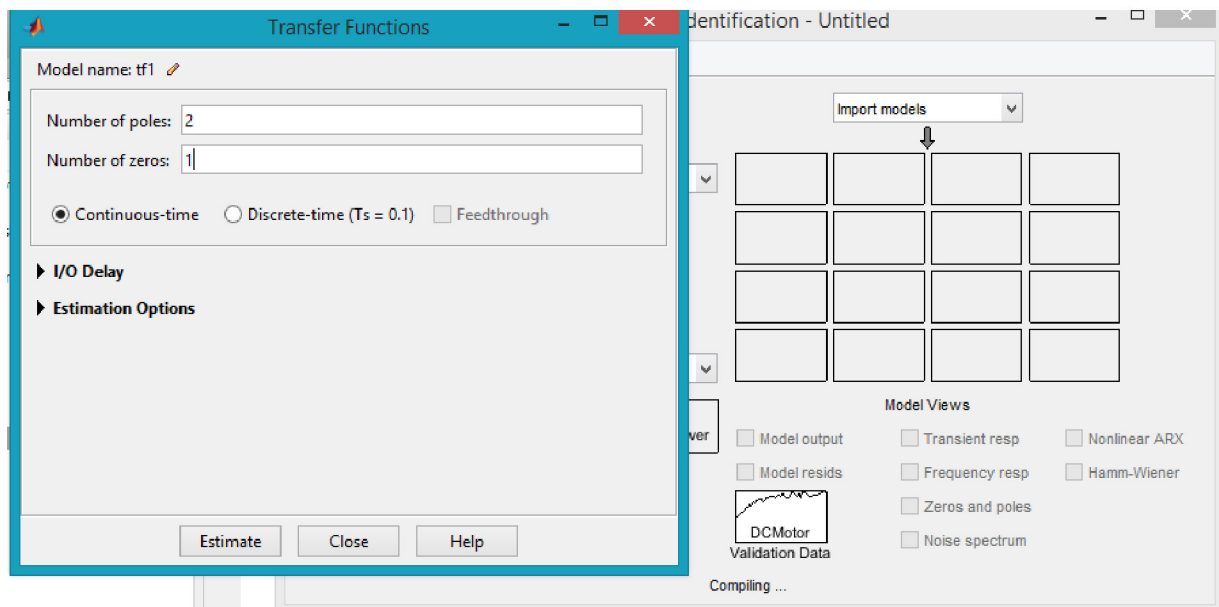


- 4) After get the data we must determine the estimation model from the choice estimate as the following figure.



We can choose the models from menu , and we study the transfer function models

5) The estimation window will appear and can be determine the order of estimated model

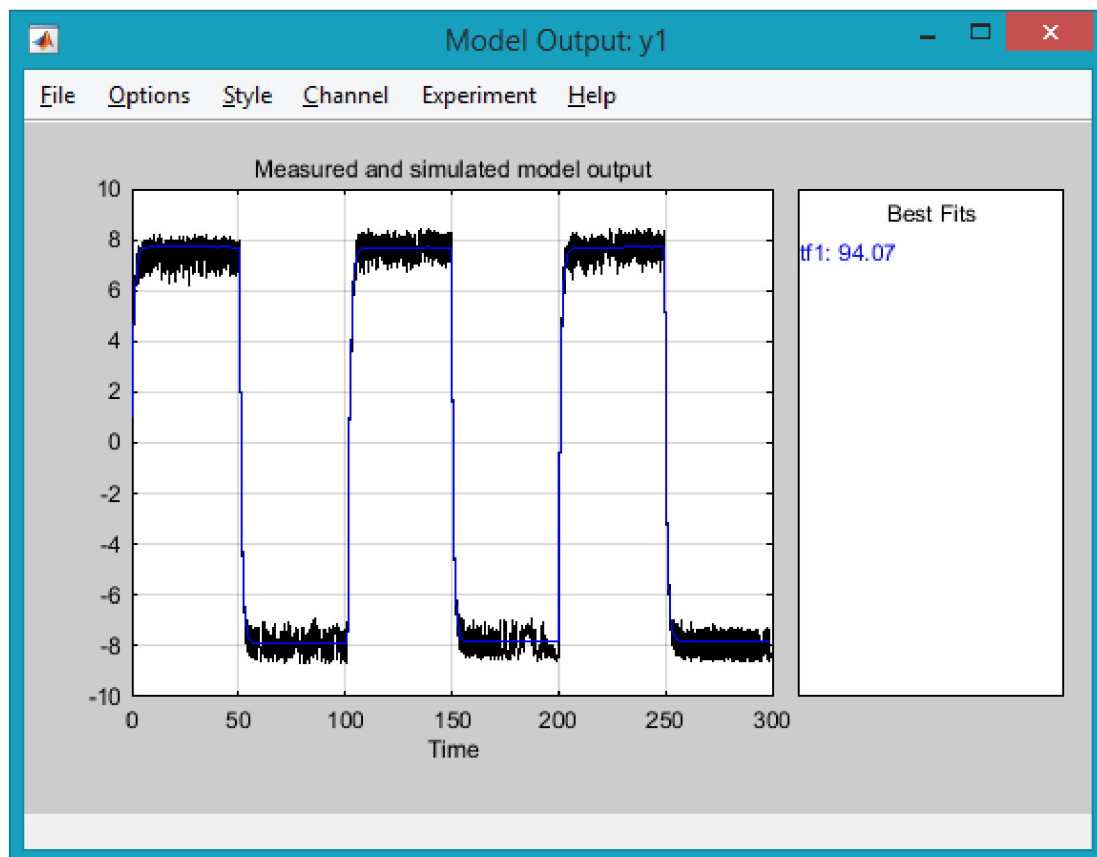
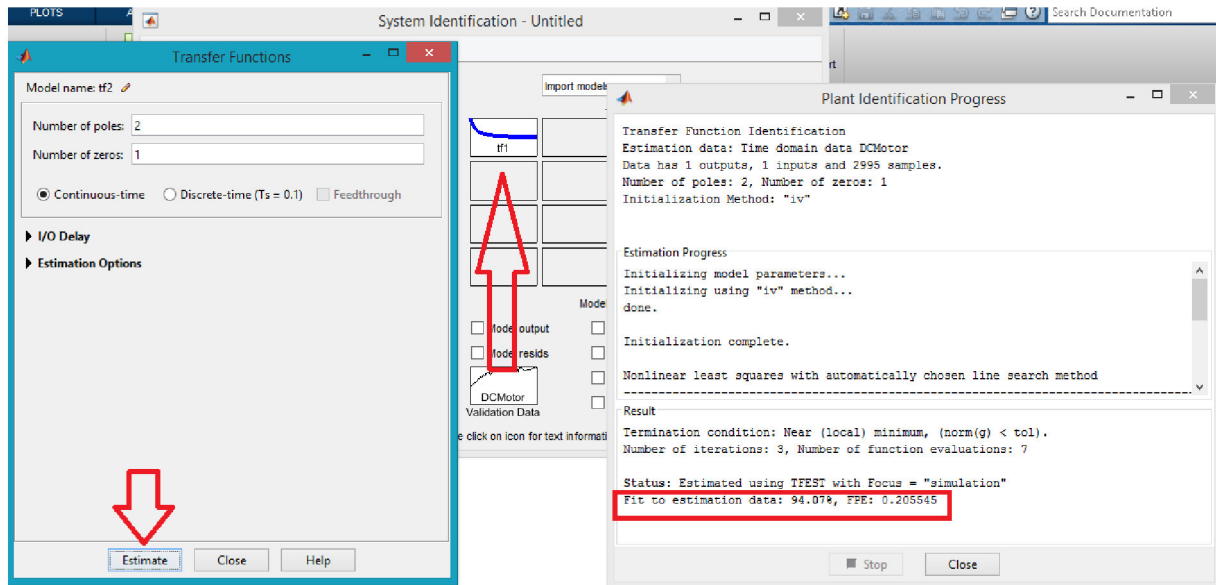


Determining the number of poles and zeros depends on your knowledge about the system, or by trial and error and choose the model that gives you the most fit value and minimum error.

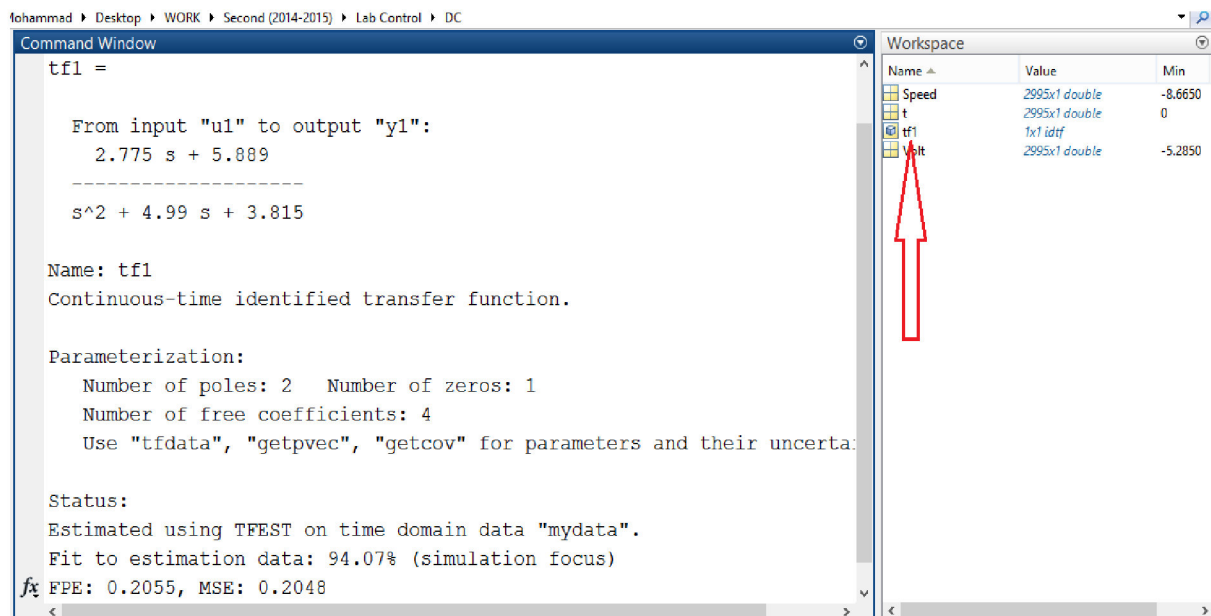
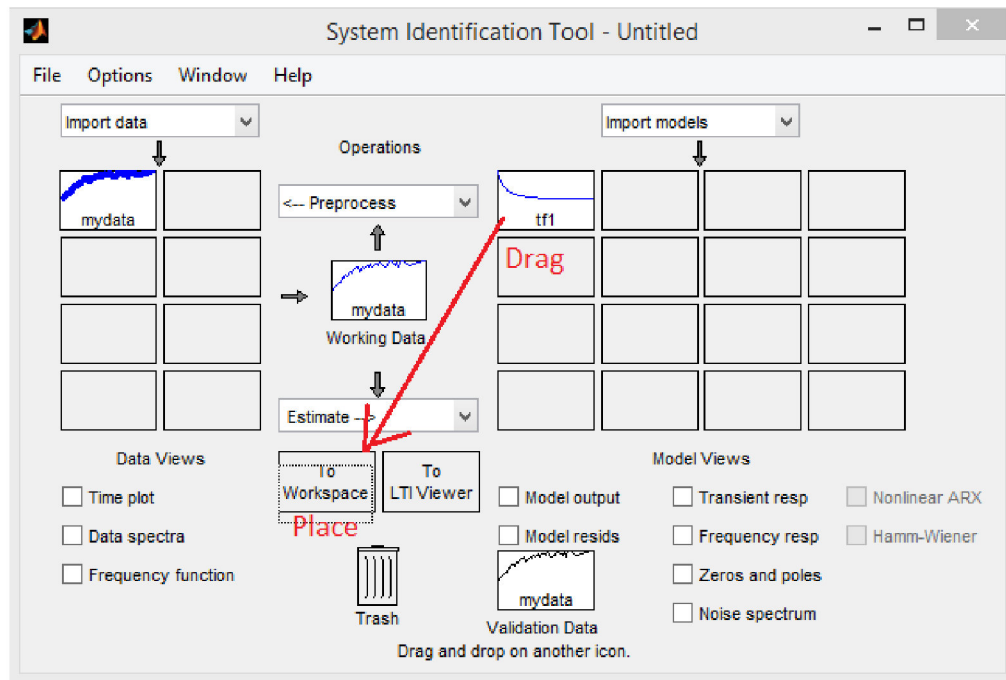
Notes:

- Start with small value of number of poles and zeros if the estimated model is not accurate, increase the number by one.
- Number of poles must be increased by one than number of zeros

After that press to estimate and the model will store in the **Import models** window. In addition, can be show the output response of model by check the **model output**



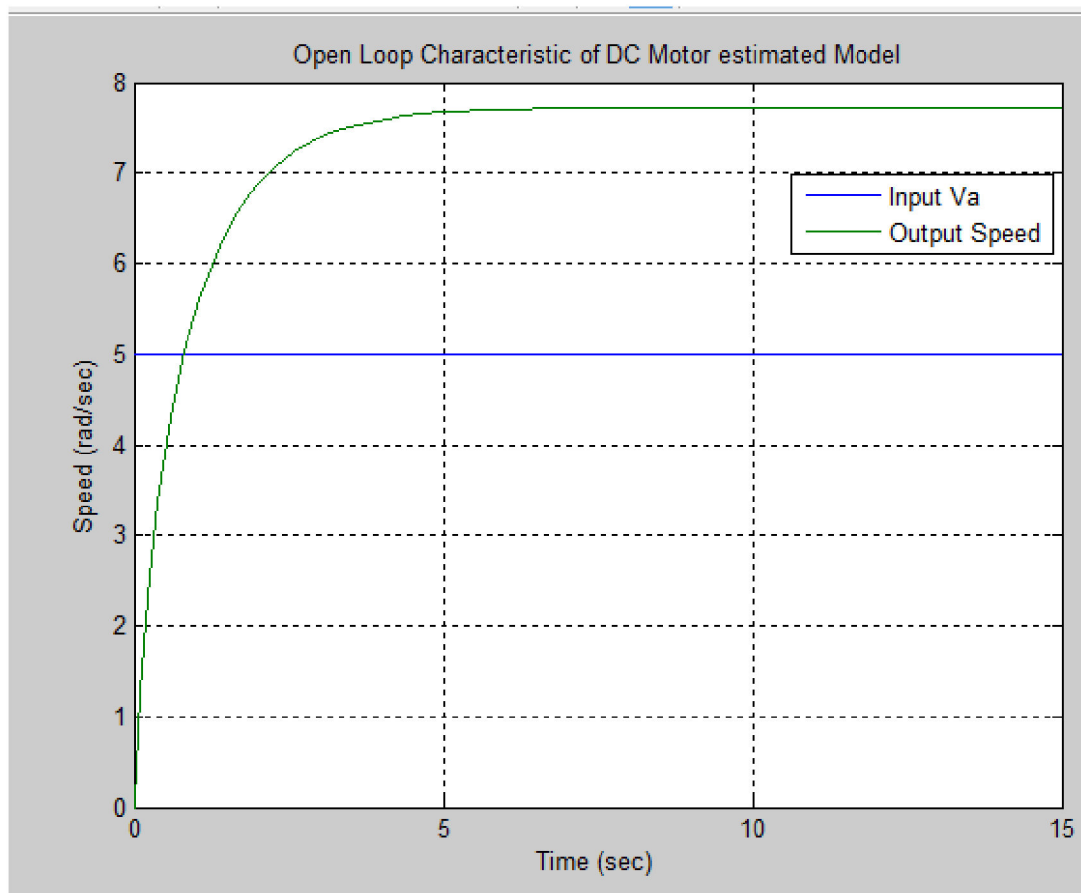
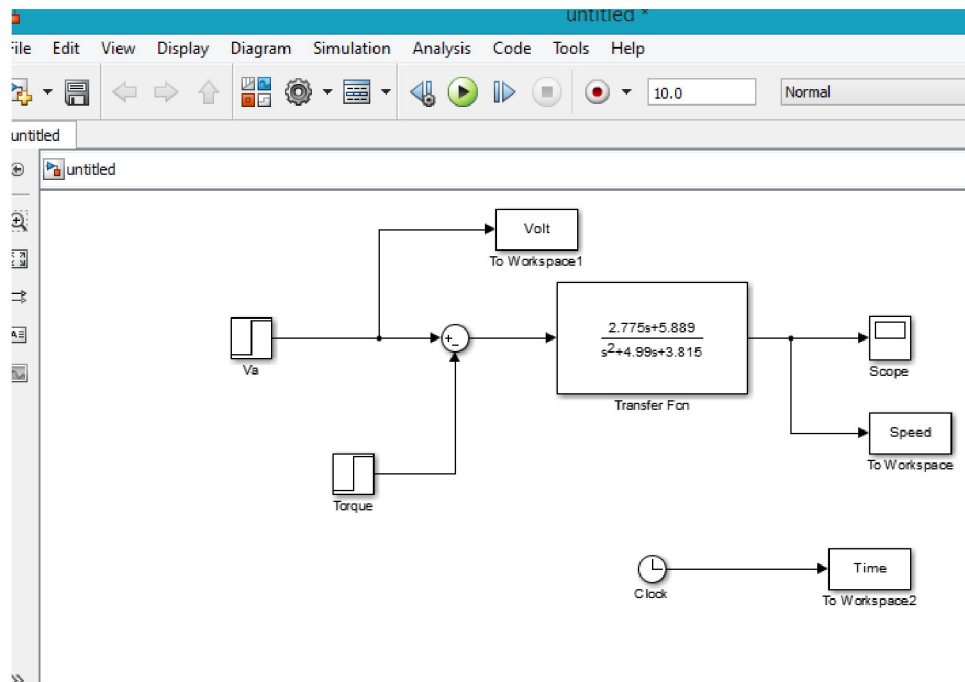
As shown in the figure the effect of noise to the response, and we can take the model to work space by drag from the *import model* and paste to *workspace*



The open loop transfer function of the DC Motor, is given by

$$G(s) = \frac{2.775s + 5.889}{s^2 + 4.99s + 3.815}$$

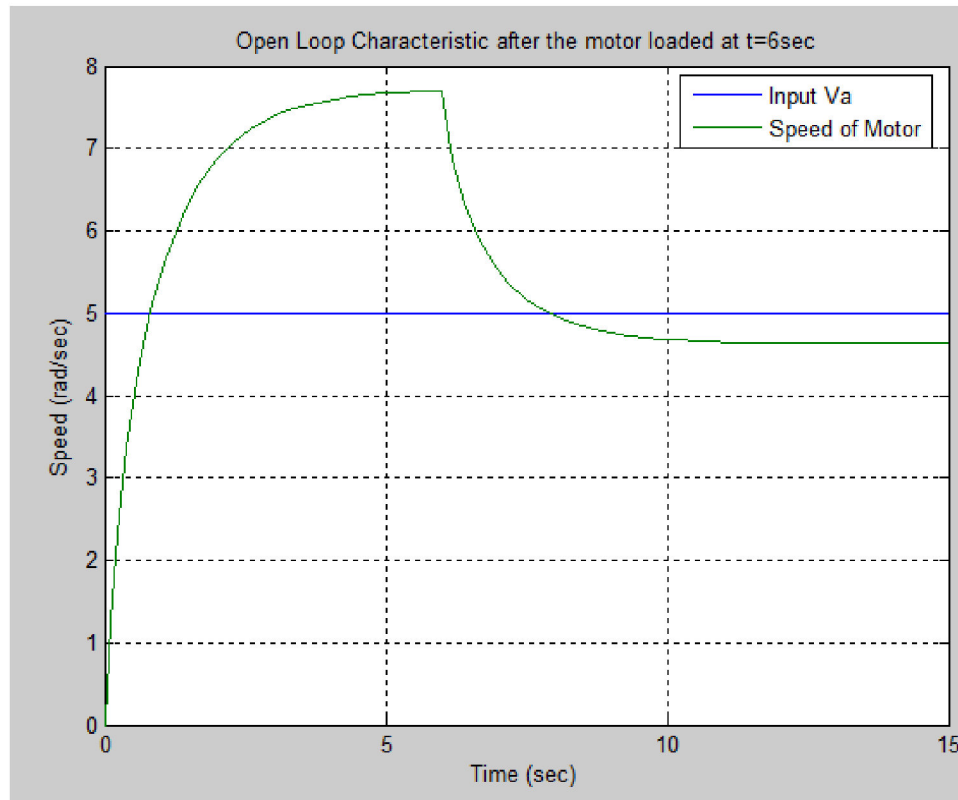
No can be simulate using Simulink



- **The effect of adding a Load to DC Motor**

Assume the DC motor that was testing in the section 6 is loaded after 6 second from the starting time by constant value = 2

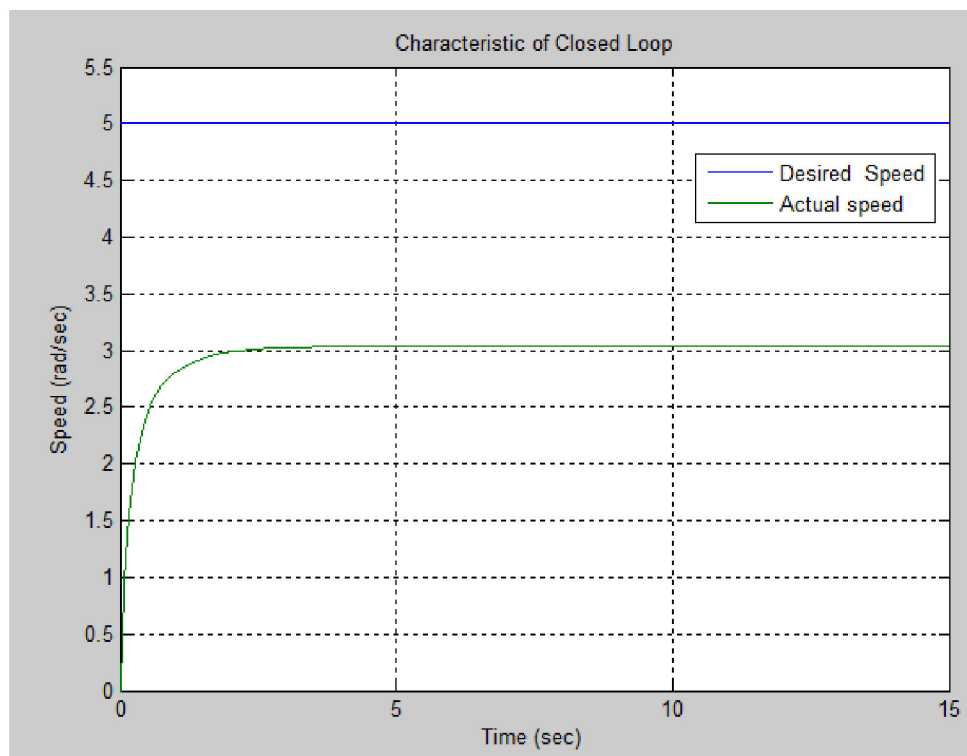
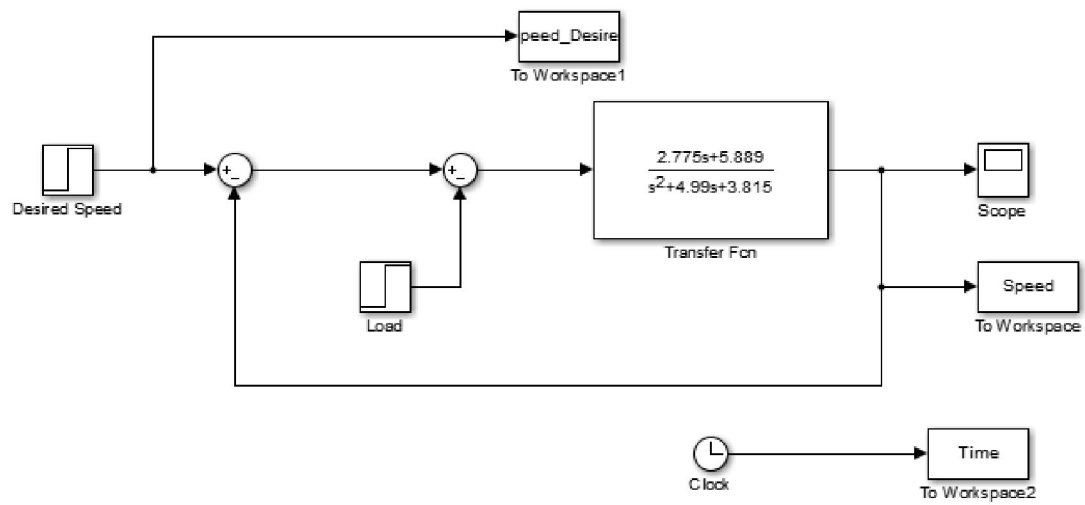
The response of open loop system is shown in the following figure



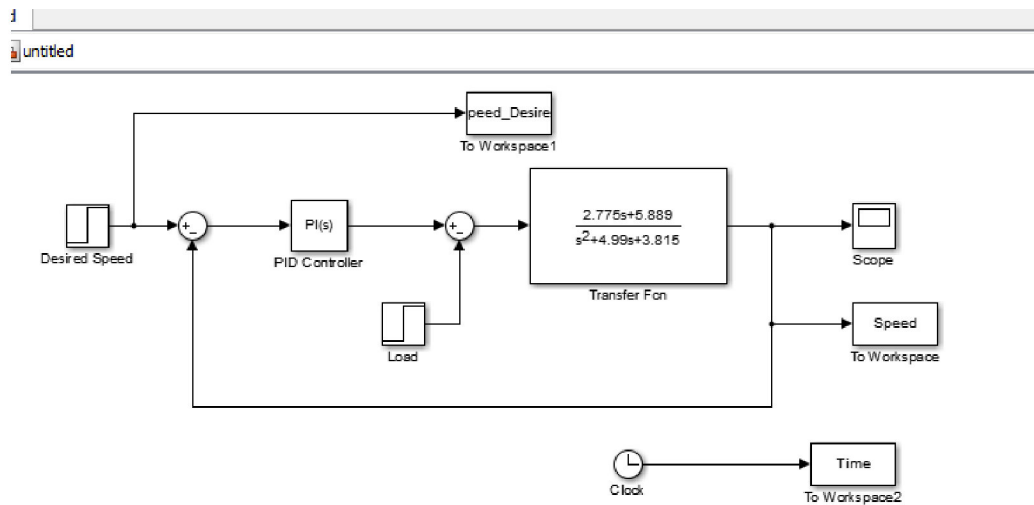
As shown in the figure the speed of motor was decreased after the torque load applied and the motor is run at different speed.

Now will design PID as the step was discussed in the previous section and study the speed characteristic of DC motor without load and under load.

The Simulink block diagram of closed loop system



The Simulink block after add the controller to system



PI Controller Parameter

Time domain:

☒ Continuous-time

☐ Discrete-time

Main PID Advanced Data Types State Attributes

Controller parameters

Proportional (P): 2.03077437916113 [Compensator formula](#)

Integral (I): 7.34202629389253

[Tune...](#)

$P + I \frac{1}{s}$

Initial conditions

Source: internal

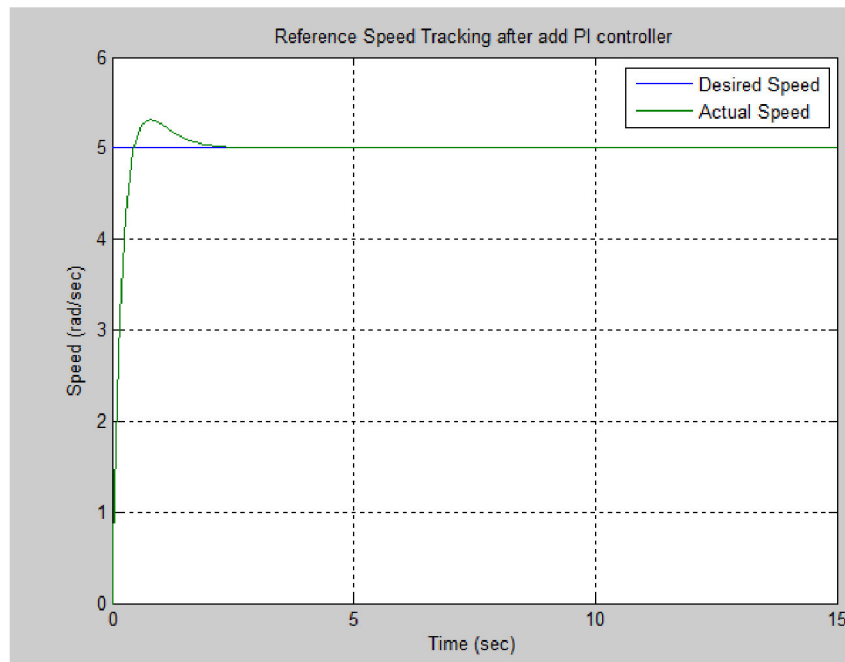
Integrator: 0

External reset: none

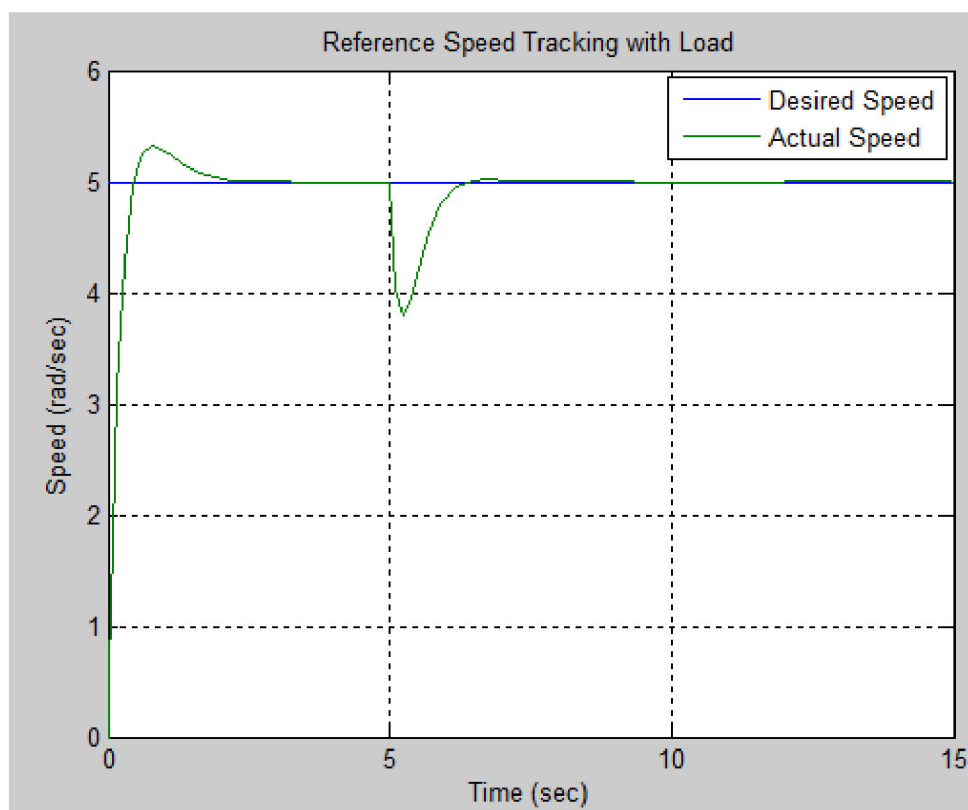
☐ Ignore reset when linearizing

☒ Enable zero-crossing detection

OK Cancel Help Apply



Now assume the motor was loaded after 5 sec by the constant value = 5 N



References

- 1) http://www.profjrwhite.com/system_dynamics/sdyn/s6/s6mathm/s6mathm.html
- 2) <http://www.mathworks.com/products/sysid/>